

Piecewise Functions Card Sort

Student Answer Sheet

Match each question card with an answer card.

Question Cards

Q1: For $f(x) = \begin{cases} 5x - 1, & x < -2 \\ -x + 6, & x \geq -2 \end{cases}$, find the sum of $f(-6)$ and $f(5)$.	Q2: For $f(x) = \begin{cases} x^2 - 10, & x \leq 5 \\ x - 8, & x > 5 \end{cases}$, find the sum of $f(-1)$ and $f(-5)$.
Q3: Graph $f(x) = \begin{cases} 3 + x, & x < -2 \\ -2x - 3, & x \geq -2 \end{cases}$.	Q4: Graph $f(x) = \begin{cases} x + 2, & x < -2 \\ 2x + 1, & x \geq -2 \end{cases}$.
Q5: Graph $f(x) = \begin{cases} -2x + 3, & x < 0 \\ 3x - 2, & x \geq 0 \end{cases}$.	Q6: Graph $f(x) = \begin{cases} 2x + 3, & x \leq -2 \\ \frac{1}{4}x + 5, & x > -2 \end{cases}$.
Q7: Given the graph below, find $f(-1) + f(-3)$.	Q8: Given $f(x) = \begin{cases} 2x - 3, & x \leq 0 \\ \frac{1}{3}x, & 0 < x < 5 \\ x + 1, & x \geq 5 \end{cases}$, find $f(-4) + f(0) + f(9)$.
Q9: Graph $f(x) = \begin{cases} 3x, & x \neq 1 \\ 5, & x = 1 \end{cases}$.	Q10: Write the equation for the piecewise function whose graph is shown.
Q11: Write the equation for the piecewise function whose graph is shown.	Q12: Write the equation for the piecewise function whose graph is shown.
Q13: Write the equation for the piecewise function whose graph is shown.	Q14: Graph $f(x) = \begin{cases} 1 + x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$.
Q15: Graph $f(x) = \begin{cases} 3 + x, & x < 0 \\ x^2 - 1, & x \geq 0 \end{cases}$.	Q16: Graph $f(x) = \begin{cases} 2x + 1, & x \leq -1 \\ x^2 - 2, & x > -1 \end{cases}$.

Answer Cards

A1:	A2:	A3: 3
A4: -2	A5: $f(x) = \begin{cases} x+2, & x < -1 \\ -x+4, & -1 < x < 1 \\ -x+4, & x \geq 1 \end{cases}$	A6: -17.5
A7:	A8: $f(x) = \begin{cases} x, & x < 0 \\ 2x, & x > 0 \end{cases}$	A9:
A10: $f(x) = \begin{cases} x, & x \leq -1 \\ x+3, & -1 < x < 1 \\ x+1, & x \geq 1 \end{cases}$	A11: -25	A12:
A13:	A14:	A15:
A16: $f(x) = \begin{cases} 3, & x < -1 \\ x^2+2, & x < -1 \\ x-1, & x \geq -1 \end{cases}$		